Project theme Discover the Wonders of AI

How to group? Heterogeneous grouping was employed to randomly assign 50 students into 10 teams of 5. Factors such as gender and prior IT skills were balanced to ensure diversity within each group, fostering collaborative learning.

Activity examples Campus Flora Guardians: AI-Human Bloom Quest

Learning objectives

①Apply CT skills: Decompose plant identification and evaluate different methods (manual vs. AI) to understand the logic behind computer vision.
②Enhance learning engagement: Boost behavioral, cognitive, and emotional engagement through authentic, engaging campus exploration.
③Master subject knowledge: Learn the classification and characteristics of flora, and understand the principles and limitations of AI image recognition.

Tasks & Activity flow

①Identify Manually: Use traditional plant guides to make preliminary identifications based on features like shape, color, and venation.
②Identify with AI: Use the image recognition feature of the course assistant to identify the same plants.
③Compare & Analyze: Compare the results, confidence levels, and additional information from both methods, and analyze discrepancies and their reasons.
④Create & Share: Produce a simple "Campus Plant Identification Report" and share the most interesting finding from the quest.

Personalized recommended exercises

Tier 1: Foundational Reinforcement (Completed basic identification tasks but struggled with comparative analysis and understanding underlying principles. AI is a tool.)
①List the names of the three plants you have successfully identified and 1-2 key identification features. ②Briefly describe the basic steps our AI course assistant likely takes to identify a flower from an image.
Tier 2: Skill Advancement (Successfully completed the task and performed basic comparison between manual and AI identification. Beginning to analyze AI's strengths and limitations.)
①Compare the main differences between manual identification and AI image recognition in terms of accuracy, efficiency, and source of information. ②In what scenarios during the activity was AI identification more reliable than manual identification? When was it potentially less reliable? Please provide examples.
Tier 3: Innovative Extension (Conducted in-depth analysis and posed insightful questions about AI's technical details, ethics, or future applications.)
①Some argue that over-relying on AI diminishes our ability to observe and explore the world firsthand. Based on your experience, what is your perspective on this viewpoint?
②Design a simple technical concept for a "Smart Campus Botanical Garden." Explain how you would integrate sensors, AI, and our course assistant to enhance the visitor experience.

Classroom implementation snapshots